

Numerical Analysis Project: Air Quality Prediction

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1 Introduction of Air Pollution

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Air pollution can cause health effects. Short-term health effects include the harm of eye, nose, and throat, headaches, allergic reactions, and upper respiratory infections. Some long-term health effects, such as lung cancer, brain damage, liver damage, kidney damage, heart disease, respiratory disease. Unexpectedly, it can even cause diabetes. The effects of air pollution for the environment are haze, eutrophication, and global climate change. The current criteria pollutants are Carbon Monoxide (CO), Lead (Pb), Nitrogen Dioxide (NO₂), Ozone (O₃), Particulate matter (PM), Sulfur Dioxide (SO₂). The Air Quality System (AQS)² contains ambient air pollution data collected by EPA, state, local, and tribal air pollution control agencies from over thousands of monitors. The methodology is that EPA regulations require state environmental agencies to report air monitoring data at least quarterly. The monitoring agencies must report the measured data, along with metadata about the site, monitoring equipment, and associated quality assurance data to the AQS. Data for a given calendar quarter are due to EPA by the end of the following quarter. Office of Air Quality Planning and Standards (OAQPS)³: Its primary mission is to preserve and improve air quality in the United States. To accomplish this, OAQPS:

1. compiles and reviews air pollution data.
2. develops regulations to limit and reduce air pollution.
3. assists states and local agencies with monitoring and controlling air pollution.
4. makes information about air pollution available to the public.
5. reports to Congress the status of air pollution and the progress made in reducing it. Since different pollutants have different effects, the National Ambient Air Quality Standards (NAAQS) are also different. Some pollutants have standards for both long-term and short-term averaging times. The short-term standards are designed to protect against acute or short-term health effects, while the long-term standards were established to protect against chronic health effects. Air quality index (AQI)⁴ which quantifies air quality in a region is used to express how polluted the air is currently or how polluted it is forecasted to become. There are five steps of AQI:

1. When AQI is between 0-50, the air pollution level is excellent.(Green)
2. When AQI is between 51-100, the air pollution level is good.(Yellow)
3. When AQI is between 101-150, the air pollution level is lightly polluted.(Orange)
4. When AQI is between 151-200, the air pollution level is moderately polluted.(Red)
5. When AQI is between 201-300, the air pollution level is heavily polluted.(Purple)
6. When AQI is over 301, the air pollution level is severely polluted.(Dark Red)

2 Introduction of domestic AQI and International AQI

The International Air Quality Index:

The Air Quality Index (AQI) is calculated based on monitoring data for various pollutants in the air on a given day. These pollutants include ozone (O₃), fine particulate matter (PM_{2.5}), particulate matter (PM₁₀), carbon monoxide (CO), sulfur dioxide (SO₂), and nitrogen dioxide (NO₂). Each pollutant is assigned a sub-index value based on its impact on human health. The AQI for a specific monitoring

¹ Air Quality Prediction: Big Data and Machine Learning Approaches International Journal of Environmental Science and Development, Vol. 9, No. 1, January 2018

² Air Quality System (AQS) <https://www.epa.gov/aqs>

³ About the Office of Air and Radiation (OAR) (<https://www.epa.gov/aboutepa/about-office-air-and-radiation-oar>)

⁴ Air Quality index (<https://airtw.moe.gov.tw/CHT/Information/Standard/AirQualityIndicator.aspx>)

station on a given day is determined by the highest sub-index value among these pollutants. The calculation of AQI among different countries is essentially same and based on this formula⁵ :

$$IAQI_P = \frac{IAQI_{Hi} - IAQI_{Lo}}{BP_{Hi} - BP_{Lo}}(C_P - BP_{Lo}) + IAQI_{Lo}$$

$IAQP_P$ = the Air Quality Sub-Index for the respective region for pollutant parameter P,

C_P = the pollutant concentration of pollutant parameter P in the respective region,

BP_{Hi} = the concentration breakpoint of pollutant parameter P that is $\leq C_P$,

BP_{Lo} = the concentration breakpoint of pollutant parameter P that is $\geq C_P$,

$IAQI_{Hi}$ = the index breakpoint corresponding to BP_{Hi} ,

$IAQI_{Lo}$ = the index breakpoint corresponding to BP_{Lo} ,

with the only difference being in the index breakpoint may be different. For example⁶:

臺灣空氣品質指標 (AQI) 計算方式如下：

各測項即時濃度依下列公式計算後，再對應下表得出 O_3 、 $PM_{2.5}$ 、 PM_{10} 、 CO 、 SO_2 、 NO_2 等 6 個測項之即時副指標值，再取出其中最大值為即時空氣品質指標，該最大值測項即為指標污染物：

$O_{3,8h}$ ：取最近連續 8 小時移動平均值 (例如今日上午 10 點發布的 O_3 的 8 小時濃度平均值，是取今日上午 2 點至上午 9 點監測數據的平均值。)

O_3 ：取即時濃度值

$PM_{2.5}$ ： $0.5 \times$ 前 12 小時平均 + $0.5 \times$ 前 4 小時平均 (前 4 小時 2 筆有效，前 12 小時 6 筆有效)

PM_{10} ： $0.5 \times$ 前 12 小時平均 + $0.5 \times$ 前 4 小時平均 (前 4 小時 2 筆有效，前 12 小時 6 筆有效)

CO ：取最近連續 8 小時移動平均值 (例如今日上午 10 點發布的 CO 的 8 小時濃度平均值，是取今日上午 2 點至上午 9 點監測數據的平均值。)

SO_2 ：取即時濃度值

$SO_{2,24h}$ ：取最近連續 24 小時濃度平均值 (例如今日上午 10 點發布的 SO_2 的 24 小時濃度平均值，是取前 1 天上午 10 點至今日上午 9 點監測數據的平均值。)

NO_2 ：取即時濃度值

⁵Technical Assistance Document for the Reporting of Daily Air Quality —the Air Quality Index (AQI) . U.S. Environmental Protection Agency. May 2016. EPA-454/B-16-002.

⁶空氣品質指標. (n.d.). 環境部空氣品質監測網.(<https://airtw.moenv.gov.tw/>)

3 Sliding Window

3.1 Introduction

The sliding window feels more like a technical application tool that can be used for data processing, especially input. Fixing the throughput of the data by the size of the window, rather than dropping it one by one, it drops it in packets, which solves the problem of traffic congestion. There is a paper⁷ on predicting the next week's stock prices using sliding windows, so it should be possible to apply it to our project as long as we have a prediction algorithm.

3.2 Operation

Figure 1 illustrates a simple example of how a sliding window works. It will start by inputting a new data from the right side of the window, and remove the oldest data, which is the leftmost one in the window. (The start and end positions of a window are typically controlled by two pointers.)

Figure 2 shows the moving averages of gold stock prices with different time periods. The orange line is 5 days, the purple line is 10 days, the blue line is 30 days and the pink line is 60 days. It is obvious that the longer the time period, the smoother the line becomes.

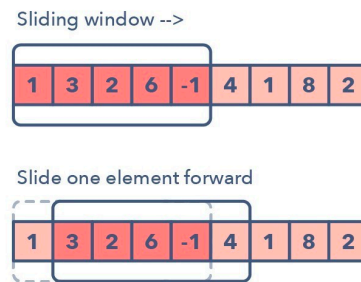


Figure 1: operation process ⁸



Figure 2: Moving Average (MA) with different time period ⁹

⁷Link of paper: <https://hdl.handle.net/11296/xtbfeq>

⁹Reference: <https://www.linkedin.com/pulse/sliding-window-two-pointer-technique-saiful-islam-rasel>

⁹Reference: <https://reurl.cc/4W9RLV>

4 論文回顧

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4.1 LightGBM

A trustworthy tool for implementing gradient boosting in decision trees is LightGBM (Zhou et al., 2022). LightGBM uses tree-based learning strategies, making it a suitable choice for gradient boosting. It provides quicker training and higher production thanks to its decentralised and effective architecture. A histogram-based method called LightGBM conducts variable bucketing, which improves training speed and accuracy while using less memory. When being trained, it works more quickly and can handle large and complex datasets. Support for both parallel and GPU-based learning. The method enables one to infer details about a target Y given only X as input when used in supervised learning scenarios. The LightGBM technique uses a supervised training set (X) and a loss function $L(y, f(x))$ whose predicted value is to be reduced to accomplish this $\hat{f}(x)$.

$$\hat{f} = \arg \min_f E_{y,x} L(y, f(x))$$

4.2 Random forest

Random forest regression is one form of machine learning-based regression technique (Ganesh et al., 2021). The foundation of this tactic is a combination of bagging and random subspace approaches. Following the bagging of numerous learning trees, an ensemble method is used to merge these trees into a single forecast. From the entire collection of N examples used for training (D), n random instances are chosen to create a bootstrap sample (D_b). It is acceptable to substitute examples when creating bootstrap samples. The input vector x is used to create K distinct regression trees for the bootstrap samples. When performing regression tasks, the random forest prediction is made by averaging K predictions from regression trees $h_k(x)$.

4.3 CatBoost

A development of the gradient boosting and decision tree frameworks is CatBoost (Zhang et al., 2020). Boosting is predicated on the idea that numerous, relatively weak models can be joined to create a single, fiercely competitive prediction model that outperforms chance by a slight margin. Gradient boosting decreases errors by fitting a series of decision trees, each of which gains knowledge from the mistakes of the preceding iteration. This process of introducing new functions into the mix is repeated until the chosen loss function is no longer minimised. CatBoost's method for creating decision trees differs from traditional gradient boosting models. Instead, CatBoost creates "oblivious trees," which are made possible by requiring that all nodes at the same level test the same predictor under the same conditions. This allows for the computation of a leaf's index using just bitwise operations. By arbitrary ordering the components of D using a random permutation, π , CatBoost chooses the data to be utilised for fitting h_{t+1} . $D_k = x_1, x_2, \dots, x_{k-1}$ where $x_1, x_2, \dots, x_{k-1}$ are the elements of D arranged by the random permutation and (k) is the k th element of D under permutation, π . Instead of strictly adhering to Eq. (3), CatBoost uses a variation of it in its analysis to define the encoded value, $\hat{x}_{ik}$ for the i th categorical value during Decision Tree h_{t+1} fitting.

¹⁰Air quality prediction by machine learning models: A predictive study on the indian coastal city of Visakhapatnam

4.4 Adaptive boosting (AdaBoost) regressor

AdaBoost (Mishra et al., 2020) one of the first boosting algorithms employed, was used to address difficulties. AdaBoost modifies the data of each training sample (x_i, y_i) by applying a weight $w_1, w_2, \dots, w_N$. The fundamental learner gives each observation the same amount of thought in the initial stage. Once weights have been assigned to each observation, the weak learner may be utilised for prediction. In this method, the predictions made by the base learner after the weak learner are more likely to be accurate. This process will be carried out again until the t th iteration, after which the T_t base learning algorithm's limit will be reached. The outputs of weak learners can be combined to generate more robust learners, which enhance the ability to predict outcomes.

4.5 Extreme gradient boosting (XGBoost)

Boosting is a technique for combining numerous weak classifiers into a single effective one. Gradient Boosting served as the basis for the development of the technique known as XGBoost (Mahesh et al., 2022). Gradient Boosting's XGBoost variation outperforms the original in terms of computing efficiency, scalability, and generalisation performance. Data organization is of utmost importance while utilising XGBoost. All category data will be transformed into their numeric equivalents because XGBoost only takes numeric vectors as input. This encoding change can be made with a single hot encoding. The feature engineering and data purification stages come next. We can obtain the estimated model by using the universal function, as indicated by the following formula.

5 Machine learning¹¹

5.1 Training and Testing

Quoting from the paper: "Empirical studies show that the best results are obtained if 80 percent of the data is used for training." and "Empirical studies show that the best results are obtained if the remaining 20 percent of the data is used for testing."

Consider a dataset of size N , denoted by $\{\mathbf{x}_i, y_i\}_{i=1}^N$. For each $i \in \{1, 2, \dots, N\}$,

$$\mathbf{x}_i = (x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(D)}), x_i^{(j)} \in \mathbb{R}$$

is called a feature vector with D -dimension and $y_i \in \mathbb{R}$ is called a real-valued target.

Definition 5.1.1 A feature vector is a vector that for each $i \neq j$, $\mathbf{x}_i^{(n)}$ and $\mathbf{x}_j^{(n)}$ have the same character of different data, where $n \in \{1, \dots, D\}$.

Definition 5.1.2 A target y_i represent the classification of data $\mathbf{x}_i$

Definition 5.1.3 A loss function is to evaluate. 真實值跟預測值之間的差距

5.2 Linear Regression

Now, for a given $\mathbf{x}$, we desire to predict the most accurate y . That is, we want to use some known data examples to estimate the relationship between $\mathbf{x}$ and y .

In linear regression, we use the model of linear function to predict y .

Consider a linear function given by

$$f_{\mathbf{w},b}(\mathbf{x}) = \mathbf{w}\mathbf{x} + b, \mathbf{w} \in \mathbb{R}^D, y \in \mathbb{R},$$

and solve for $\mathbf{w}^*$ and b^* by minimizing the following equation which is also the loss function of linear regression,

$$\frac{1}{N} \sum_{i=1}^N (f_{\mathbf{w},b}(\mathbf{x}_i) - y_i)^2.$$

Then we can estimate the relationship between $\mathbf{x}$ and y by the linear function.

The reason of using binary function rather than absolute value function, which also can calculate the distance between the real value and the prediction is because it's is more easy to find the minimum value by differentiating the function.

However, the absolute value function can not be differentiate in some point.

Using higher-order function might fit the example very well. However, if there are some noise in the example, the prediction might be very far from the real value as the graph show as following:

¹¹Andriy Burkov(2019).The Hundred-Page Machine Learning Book. ISBN-13: 978-1999579500

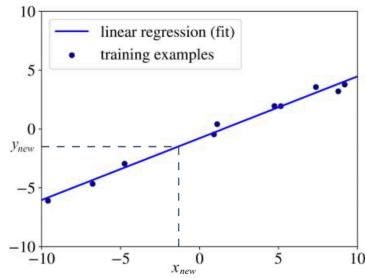


Figure 3: Linear regression of degree 2

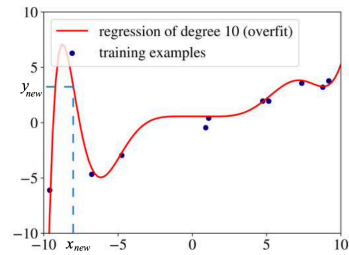


Figure 4: linear regression of degree 10

5.3 Logistic Regression

In logistic regression, for each $i \in \{1, 2, \dots, N\}$, y_i is either 0 or 1. Define $y = 0$ if y is negative, $y = 1$ otherwise.

Example:

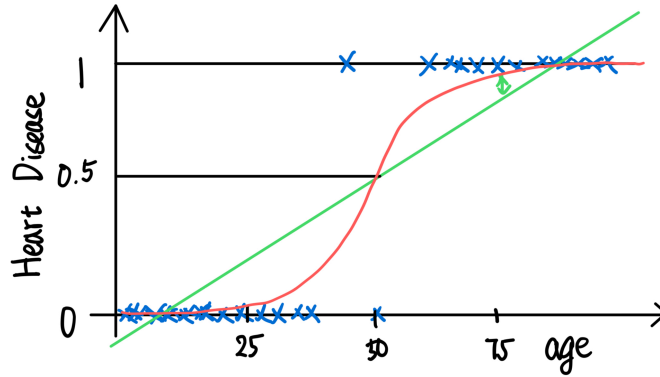


Figure 5: Example of logistic regression

In Figure 5 is about the age of people having heart disease (blue crosses). We can find out that the linear regression can not fit well in this example, there are many gaps between the data and the prediction. While the red line fits better which has function looks like $y = \frac{1}{1 + e^{-x}}$. We can observe that if $x < 0$ implies $y < 0.5$, $y > 0.5$ otherwise.

Graph:

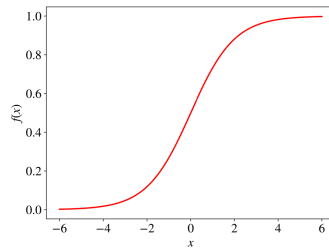


Figure 6: Graph of $y = \frac{1}{1 + e^{-x}}$

So base on the example above, for a given $\mathbf{x}$, we desire to predict the probability of y being positive. We consider the model given by

$$f_{\mathbf{w},b}(\mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}\mathbf{x}+b}}$$

and solve $\hat{\mathbf{w}}$ and $\hat{b}$ by maximize the following likelihood function,

$$L_{\mathbf{w},b} = \prod_{i=1}^N f_{\mathbf{w},b}(\mathbf{x}_i)^{y_i} (1 - f_{\mathbf{w},b}(\mathbf{x}_i))^{(1-y_i)}$$

To calculate this likelihood function, we will like to take nature log on both side, which called log-likelihood function as following:

$$l_{\mathbf{w},b} = \sum_{i=1}^N y_i \ln f_{\mathbf{w},b}(\mathbf{x}_i) + (1 - y_i) \ln(1 - f_{\mathbf{w},b}(\mathbf{x}_i))$$

which looks similar to logistic regression's loss function as following:

$$Cost = -\frac{1}{N} \sum_{i=1}^N y_i \ln f_{\mathbf{w},b}(\mathbf{x}_i) + (1 - y_i) \ln(1 - f_{\mathbf{w},b}(\mathbf{x}_i))$$

Since there is a negative sign in front of the formula, so we can consider the cost function by observing the image of $-\ln x$ as following:

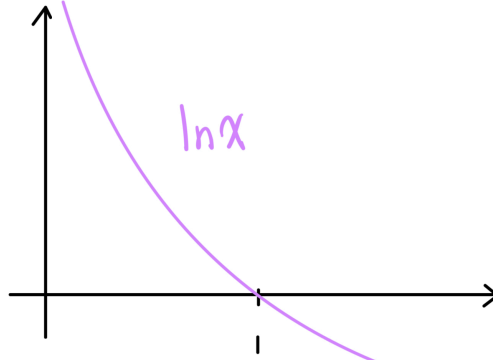


Figure 7: Graph of $-\ln x$

From Figure 7, we can know that if $y_i = 0$, when the prediction $f_{\mathbf{w},b}(\mathbf{x}_i)$ is close to zero, then $-\ln(1 - f_{\mathbf{w},b}(\mathbf{x}_i))$ in the cost function will close to zero either.

On the other hand, if the prediction $f_{\mathbf{w},b}(\mathbf{x}_i)$ is close to 1, which is far from the real value $y_i = 0$, then $-\ln(1 - f_{\mathbf{w},b}(\mathbf{x}_i))$ in the cost function will approach infinite. The case of $y_i = 1$ similarly. Which match the meaning of loss function.

5.4 Decision Tree

A decision tree is a acyclic graph that can be used to make decision. In every branching node of the graph has a specific value k , which examined $\mathbf{x}_i^{(j)}$ for some $j \in \{1, 2, \dots, D\}$, where $i \in \{1, 2, \dots, N\}$. If $\mathbf{x}_i^{(j)} < k$, then the left branch will be followed, otherwise, the right side will be followed.

There are many decision tree algorithms, we only consider the algorithm called **ID3**.

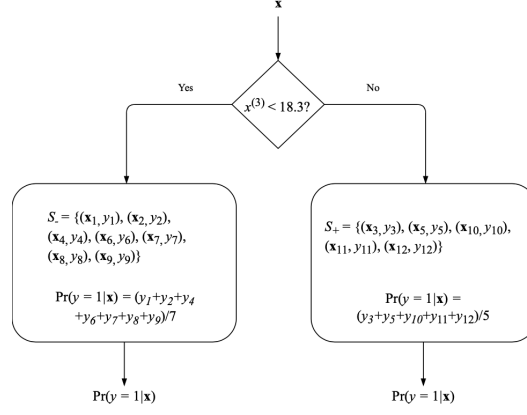


Figure 8: Decision tree

Graph: First, we let $f_{ID3} \stackrel{def}{=} Pr(y = 1|\mathbf{x})$ and $S = \{(\mathbf{x}_i, y)\}_{i=1}^N$.
We construct the model:

$$f_{ID3}^S \stackrel{def}{=} \frac{1}{|S|} \sum_{(\mathbf{x}, y) \in S} y$$

which have the same output for any $\mathbf{x}$ in S .

Then we want to split S into two sets $S_- = \{(\mathbf{x}, y) | (\mathbf{x}, y) \in S, x^{(j)} < k\}$ and $S_+ = \{(\mathbf{x}, y) | (\mathbf{x}, y) \in S, x^{(j)} \geq k\}$, which is the best split evaluated by the entropy function as following:

$$H(S) = -f_{ID3}^S \ln f_{ID3}^S - (1 - f_{ID3}^S) \ln(1 - f_{ID3}^S)$$

which attains it's maximum when S is equiprobable (有一半的例子 $y = 1$) , attains it's minimum when there only have one value($y = 1$ for all $(\mathbf{x}, y)$, or $y = 0$ for all $(\mathbf{x}, y)$)

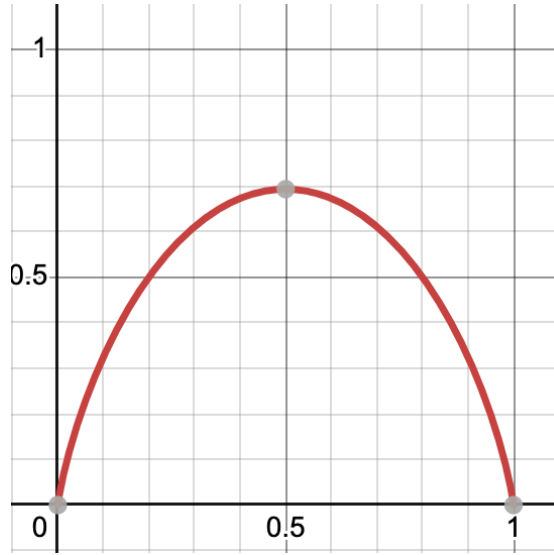


Figure 9: Entropy function

When we split a example set by some j, k , we have a entropy function of the split define as following

$$H(S_-, S_+) = \frac{|S_-|}{|S|} H(S_-) + \frac{|S_+|}{|S|} H(S_+). \quad (1)$$

In conclusion, when ever we decide a node of decision tree, we find the split such that by minimizing equation 1. Moreover, we decide which component to split by watching the ones which hasn't be picked up before and has the minimum value in equation 1 with it's minimum k .

Stopping Condition:

1. Every leaf node have the same value y .
2. Cannot find a attribute to split the example.
3. The split entropy less then a given ε .
4. The tree reaches the maximum depth given by d .

5.5 Support Vector Machine (SVM)

In SVM, for each $i \in \{1, 2, \dots, N\}$, y_i is either -1 or 1 .

We want to find a hyperplane h such that if $y_i \neq y_j$, then $(\mathbf{x}_i, y_i)$ and $(\mathbf{x}_j, y_j)$ lies on different side of h . The hyperplane h is given by two parameters $\mathbf{w}$ and b as following:

$$h : \mathbf{w}\mathbf{x} + b = 0$$

where $\mathbf{w} \in \mathbb{R}^D, b \in \mathbb{R}$.

Definition 5.5.1 The hyperplane h defined by $(\mathbf{w}, b)$ splits the data if and only if $y_i(\mathbf{w}\mathbf{x}_i + b) > 0$ for all $i = 1, 2, \dots, N$.

Take

$$\delta = \min_{n=1,2,\dots,N} y_n(\mathbf{w}\mathbf{x}_n + b).$$

Re-scale the hyperplane h by $(\frac{\mathbf{w}}{\delta}, \frac{b}{\delta})$, then we have

$$\min_{i=1,2,\dots,N} y_i(\frac{\mathbf{w}}{\delta}\mathbf{x}_i + \frac{b}{\delta}) = \frac{1}{\delta} \min_{i=1,2,\dots,N} y_i(\mathbf{w}\mathbf{x}_i + b) = \frac{\delta}{\delta} = 1$$

Definition 5.5.2 The hyperplane h splits the examples if and only if the hyperplane h determine by $(\mathbf{w}, b)$ satisfies $\min_{i=1,2,\dots,N} y_i(\mathbf{w}\mathbf{x}_i + b) = 1$.

Remark 5.5.3 Definition 5.5.1 and Definition 5.5.2 are equivalent.

Remark 5.5.4 $\mathbf{w}$ is normal to the hyperplane h .

Consider $x' \in h, u = \frac{\mathbf{w}}{\|\mathbf{w}\|}$.

Then the distance between h and x is defined by

$$dist(\mathbf{x}, h) = |u(x - x')| = \frac{|\mathbf{w}\mathbf{x} - \mathbf{w}\mathbf{x}'|}{\|\mathbf{w}\|} = \frac{|\mathbf{w}\mathbf{x} + \mathbf{b}|}{\|\mathbf{w}\|} \quad (2)$$

By Definition 5.5.1 and equation 2, we have

$$dist(\mathbf{x}_i, h) = \frac{y_i(\mathbf{w}\mathbf{x}_i + b)}{\|\mathbf{w}\|} \quad (3)$$

By equation 3, we have

$$\min_{i=1,2,\dots,N} \text{dist}(\mathbf{x}_i, h) = \frac{1}{\|\mathbf{w}\|} \min_{i=1,2,\dots,N} y_i(\mathbf{w}\mathbf{x}_i + b) = \frac{1}{\|\mathbf{w}\|}$$

which is called the margin of the separating hyperplane h .

The maximum margin of the separating hyperplane h with $(\mathbf{w}^*, b^*)$ is the hyperplane with minimum weight-norm.

So, the problem that we want the machine to solve is minimize $\|\mathbf{w}\|$ subject to $\min_{i=1,2,\dots,N} y_i(\mathbf{w}\mathbf{x}_i + b) = 1$, which is equivalent to minimize $\|\mathbf{w}\|$ subject to $\min_{i=1,2,\dots,N} y_i(\mathbf{w}\mathbf{x}_i + b) \geq 1$. Which is also called Hard margin SVM.

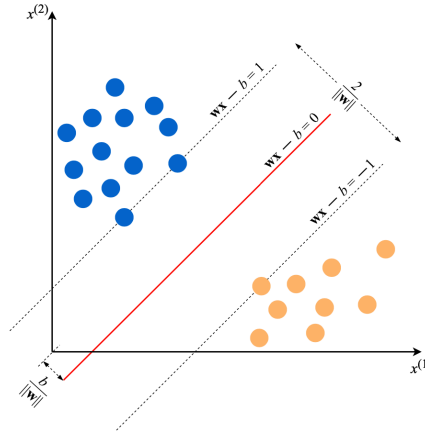


Figure 10: SVM (Hard margin SVM)

There are two questions can be ask.

1. What if there are noises such that there is no hyperplane that can separate the positive ones perfectly and the negative ones?
2. What if the dataset can not be separate by a plane, but can be separate perfectly by a high-order polynomial?

5.5.5 Dealing with noise

For the first question, we will introduce soft margin SVM.

Before showing the equation of soft margin SVM, we first introduce the hinge loss function as following:

$$\max_{i=1,2,\dots,N} (0, 1 - y_i(\mathbf{w}\mathbf{x}_i + b))$$

which have the image as following:

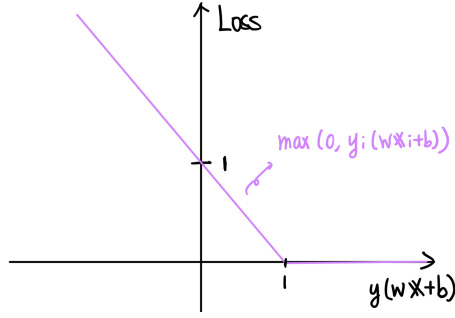


Figure 11: Hinge loss function

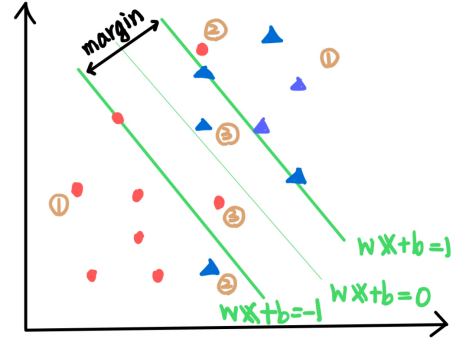


Figure 12: Soft Margin SVM

We consider three situation as in figure 12, which the red points has negative y , and the blue points have the positive y .

1. Points that lie in the correct side and outside the margin.
2. Points that lie on the wrong side.
3. Points that lie on the correct side but inside the margin.

The first case, which $y(\mathbf{w}\mathbf{x} + b) \geq 1$, we wish there are no loss, since it is correct. So, we can find out for case 1, the hinge loss function will output the value 0.

For the second case, which $y(\mathbf{w}\mathbf{x} + b) \leq 0$, we wish the loss function output a big value and the hinge function did either.

For the last case, which $0 < y(\mathbf{w}\mathbf{x} + b) < 1$. It lies on the correct side, however it is too close to $\mathbf{w}\mathbf{x} + b = 0$, so we wish it still have a small loss in this case.

After knowing the hinge loss function, we can come up the formula for soft margin SVM is to minimize the function as following:

$$\lambda \|\mathbf{w}\|^2 + \frac{1}{N} \sum_{i=1}^N \max_{i=1,2,\dots,N} (0, 1 - y_i(\mathbf{w}\mathbf{x}_i + b))$$

where the function $\frac{1}{N} \sum_{i=1}^N \max_{i=1,2,\dots,N} (0, 1 - y_i(\mathbf{w}\mathbf{x}_i + b))$ calculate the average loss of the example, and $\|\mathbf{w}\|^2$ is equivalent as hard margin SVM is to minimize $\|\mathbf{w}\|$.

The λ in front of $\|\mathbf{w}\|^2$ is a positive value such that decide rather we want a bigger margin or less noise. If λ is large, then the term of loss function will negligible. And the choose of λ is base on experience as the ϵ and d in the stopping condition of decision tree.

5.5.6 Dealing with Inherent Non-Linearity¹²

We first take a look at the example of a dataset which can not be separate by a line (hard margin SVM) as following:

¹²Radial Basis Function Kernel : Data Science Concepts, (<https://www.youtube.com/watch?v=Q0ExqOphnW0>)

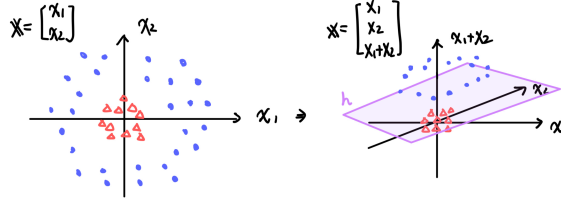


Figure 13: Example of inherent non-linear

We can see on the left side of Figure 13, the dataset although seems can be separate, however in hard margin SVM, we must only separate it by a line, which is impossible in this case.

However, we can transpose the feature vector $\mathbf{x}$ into a higher dimension vector like we do in the right side of the Figure 13. By doing this, we can separate the two sets by a plane now.

So, this is what we are going to do in this section.

By achieving this goal, we first need to do a little change on the formula of getting hard margin SVM, which is $\min \|\mathbf{w}\|^2$ such that $\forall i = 1, 2, \dots, N, y_i(\mathbf{w}\mathbf{x}_i + b) \geq 1$.

By using the method of Lagrange multiplier, $L(\mathbf{w}, b, \alpha) = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^N \alpha_i (y_i(\mathbf{w}\mathbf{x}_i + b) - 1)$, where α_i is called Lagrange multiplier.

We can get a equivalent function as hard margin SVM as following:

$$\max_{\alpha_i \geq 0} [\min_{\mathbf{w}, b} L(\mathbf{w}, b, \alpha_i)] \quad (4)$$

By calculating the min inside the brackets, we can get two equation:

$$\frac{\partial L}{\partial \mathbf{w}} = 0 \implies \mathbf{w} = \sum_{i=1}^N \alpha_i y_i \mathbf{x}_i \quad (5)$$

$$\frac{\partial L}{\partial b} = 0 \implies \sum_{i=1}^N \alpha_i y_i = 0 = \alpha^T \mathbf{y} \quad (6)$$

Note that since it is hard margin in this case, so the determine of $\mathbf{w}$ only depends on the support vectors, which are the vectors such that $y_i(\mathbf{w}\mathbf{x}_i + b) = 1$.

Base on this note, we can know that the α_i in equation 5 will only be 0 if the $\mathbf{x}_i$ is not a support vector. We keep on calculating equation 4 by plugging the method of Lagrange multiplier, we can get

$$\max_{\alpha_i \geq 0} [\min_{\mathbf{w}, b} L(\mathbf{w}, b, \alpha_i)] = \max_{\alpha_i \geq 0} \left[\sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \right] \quad (7)$$

As we say above that only support vectors(SV) have non-zero α , so we can get from equation 7

$$\max_{\alpha_i \geq 0} [\min_{\mathbf{w}, b} L(\mathbf{w}, b, \alpha_i)] = \max_{\alpha_i \geq 0} \left[\sum_{SV} \alpha_i - \frac{1}{2} \sum_{SV} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \right] \quad (8)$$

By equation 8 we can know the only information we need to know is about the pair inner product of any two support vectors. Which is called SVM dual.

The reason using SVM dual rather than hard margin SVM is because in real world, we may have not many examples(N), however every example has many character(D), such that $D \gg N$. So when we calculate hard margin SVM, we need a huge space to save a $N \times D$ matrix. However in dual SVM, the size of matrix is only $N \times N$ since we only need to save the inner product of some support vectors, which is much less than hard margin SVM.

So, as we show in Figure 13, we want to find a transform $T[x^{(1)}, x^{(2)}]^T = [1, x^{(1)}, x^{(2)}, x^{(1)}x^{(2)}, \dots]^T$ such that we can find the hyperplane of the dataset.

However, the transform $T(\mathbf{x})$ might be high dimension, in some case we even like to consider a infinite dimension.

By SVM dual we can get that for every example we want to find it's SVM, we need to find the inner product of every two support vectors as following process image:

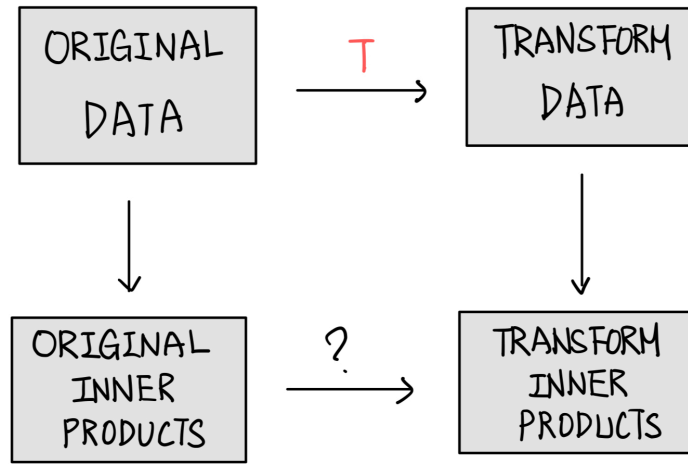


Figure 14: Process of Kernels

As Figure 14, if we do the transform T first it might be hard to do the inner product since the dimension is too large.

So we will like to go the other side of the process, which is calculate the inner product first, then figure out a solution such that have the same result.

The solution is consider a function define as following:

$$K_{poly(n)}(\mathbf{x}_i \cdot \mathbf{x}_j) = (1 + \mathbf{x}_i \cdot \mathbf{x}_j)^n = (1 + \sum_{k=1}^D x_i^{(k)} x_j^{(k)})^n \quad (9)$$

We can find out that the equation 9, which is called Kernel SVM, gives the same terms as we do the transform first and calculate the inner product.

Finally, base on above, we can define the Radial Basis Function Kernel which bring us from original inner products to transform inner products in the process above:

$$\begin{aligned}
K_{RBF}(\mathbf{x}_i \cdot \mathbf{x}_j) &= e^{-\frac{1}{2}\|\mathbf{x}_i - \mathbf{x}_j\|^2} \\
&= e^{-\frac{1}{2}(\mathbf{x}_i - \mathbf{x}_j)^T(\mathbf{x}_i - \mathbf{x}_j)} \\
&= e^{-\frac{1}{2}(\mathbf{x}_i^T \mathbf{x}_i - 2\mathbf{x}_i^T \mathbf{x}_j + \mathbf{x}_j^T \mathbf{x}_j)} \\
&= e^{-\frac{1}{2}(\mathbf{x}_i^T \mathbf{x}_i + \mathbf{x}_j^T \mathbf{x}_j)} e^{\mathbf{x}_i^T \mathbf{x}_j} \\
&= C e^{\mathbf{x}_i^T \mathbf{x}_j} \\
&= C' e^{(1 + \mathbf{x}_i^T \mathbf{x}_j)} \\
&= C' \sum_{n=1}^{\infty} \frac{(1 + \mathbf{x}_i^T \mathbf{x}_j)^n}{n!} = C' \sum_{n=1}^{\infty} \frac{K_{poly(n)}(\mathbf{x}_i \cdot \mathbf{x}_j)}{n!}
\end{aligned} \tag{10}$$

By equation 10, we get a all term of infinite dimension vector inner products by easily calculate $e^{-\frac{1}{2}\|\mathbf{x}_i - \mathbf{x}_j\|^2}$ except of actually calculate the inner product of two infinite dimension vectors.

In other word, we can consider all the interaction between $x^{(1)}, x^{(2)}$ in transformation T by calculating $e^{-\frac{1}{2}\|\mathbf{x}_i - \mathbf{x}_j\|^2}$. Which is the goal we want to achive in this section.

5.6 k-Nearest Neighbors

k-Nearest Neighbors (kNN) unlike other learning algorithm that can discard the training data after the modal was made. kNN keeps all the training data, when a previously unseen example $\mathbf{x}$ was input, kNN will find k training examples which is closest to $\mathbf{x}$, and return majority label or average label to $\mathbf{x}$'s label y .

We define the distance function here by following:

$$d(\mathbf{x}_i, \mathbf{x}_j) = \frac{\sum_{k=1}^D x_i^{(k)} x_j^{(j)}}{\sqrt{\sum_{k=1}^D (x_i^{(k)})^2} \sqrt{\sum_{k=1}^D (x_j^{(k)})^2}}$$

6 實作前置作業

6.1 Collecting dataset

First, to predict AQI, we need the real AQI data(the yellow column). On the left side of AQI, the data are those we think might affect AQI directly, such as precipitation(降水量), wind speed(風速), wind direction(風向), temperature(氣溫), relative humidity(相對溼度), ultraviolet radiation(紫外線). We collect those data from Central Weather Administration¹³. Why we collect the data on the right side of AQI is since that AQI are calculated based on those monitoring data for various pollutants. We collect those data from Ministry of Environment¹⁴.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	date_name	降水量	風速(m/s)	風向	氣溫	相對溼度(%)	紫外線	AQI	SO2	CO	PM10	NO2	O3(8hr)	PM2.5
2	2023/10/31	0	2.4	58	24.3	69	8	39	2	2	11	22	39	20
3	2023/10/30	0	1.4	52	23.5	74	6	38	5	3	14	33	38	26
4	2023/10/29	0	1.4	56	22.9	70	5	40	5	3	16	23	40	29
5	2023/10/28	14.5	1.8	53	21.9	78	4	38	2	3	10	22	38	17
6	2023/10/27	0	1.8	50	24.3	72	5	38	2	3	18	32	38	32
7	2023/10/26	0	1.5	92	25.1	68	8	80	5	5	23	38	80	49
8	2023/10/25	0	1.9	38	25.6	75	8	38						
9	2023/10/24	0	1.3	59	25.6	77	6	50	5	5	24	35	50	47
10	2023/10/23	0	1.2	87	25.8	71	8	46	5	3	14	38	46	24
11	2023/10/22	0	1.8	50	23.9	71	6	34	2	2	7	18	34	13
12	2023/10/21	0	2.9	57	22.5	72	5	37	2	3	14	18	37	22
13	2023/10/20	0	2	1	23.9	85	5	51	2	3	23	25	40	51
14	2023/10/19	0	0.9	21	25.9	76	7	169	5	6	44	28	169	94
15	2023/10/18	0	0.8	20	25.8	70	8	140	8	7	39	60	140	81
16	2023/10/17	0	1.8	32	25.4	56	8	126	2	5	32	22	126	67
17	2023/10/16	0	2.3	55	24.9	55	10	90	2	3	32	17	90	65
18	2023/10/15	0	2	58	24.1	62	8	108	5	5	30	22	108	65
19	2023/10/14	0	1.5	52	24.2	70	5	64	8	5	22	28	64	45
20	2023/10/13	0	1.8	29	25.2	68	9	54	2	3	22	28	54	29
21	2023/10/12	0	2.2	53	25.1	67	9	47	5	3	20	25	47	29
22	2023/10/11	0.5	3	58	24.6	67	10	58	2	3	18	18	58	27
23	2023/10/10	0	2.4	30	24.6	81	11	40	2	2	8	15	40	13
24	2023/10/9	0	2	53	24.5	87	3	42	0	2	12	13	42	18

6.2 Confusion Matrix

When evaluating the performance of a model, we often use confusion matrix to give a analysis. So we first define:

True positive (TP) = A test result that correctly indicates the presence of a attribute

True negative (TN) = A test result that correctly indicates the absence of a attribute

False positive (FP) = A test result which wrongly indicates that a particular attribute is present

False negative (FN) = A test result which wrongly indicates that a particular attribute is present

	Actual positive	Actual negative
Predicted positive	TP	FP
Predicted negative	FN	TN

In general, We initially use *Accuracy* to know how well our prediction models are.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

¹³Central Weather Administration (<https://www.cwa.gov.tw/V8/C/D/UVIHistory.html>)

¹⁴Ministry of Environment (https://data.moenv.gov.tw/dataset/detail/AQX_P_434)

7 機器學習結果

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 15: 使用雨量預測 AQI

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 16: 使用風速預測 AQI

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 17: 使用風向預測 AQI

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 18: 使用氣溫預測 AQI

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 19: 使用濕度預測 AQI

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 20: 使用紫外線預測 AQI

accuracy: 80.75%

	true false	true true	class precision
pred. false	38	10	79.17%
pred. true	41	176	81.11%
class recall	48.10%	94.62%	

Figure 21: 使用前一日 AQI 預測 AQI

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 22: 使用 p 預測 AQI

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 23: 使用預測 AQI

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 24: 使用預測 AQI

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 25: 使用預測 AQI

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 26: 使用預測 AQI

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 27: 使用預測 AQI

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 28: 使用預測 AQI

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 29: 使用預測 AQI

8 改善資料

8.1 Performance of Model

In the process of collecting data, it's possible to observe a significant imbalance in the quantities of positive and negative samples. We call it imbalanced dataset. It have problems that how accurately are we actually predicting both the majority and minority classes? So we need other indicator to improve the condition.

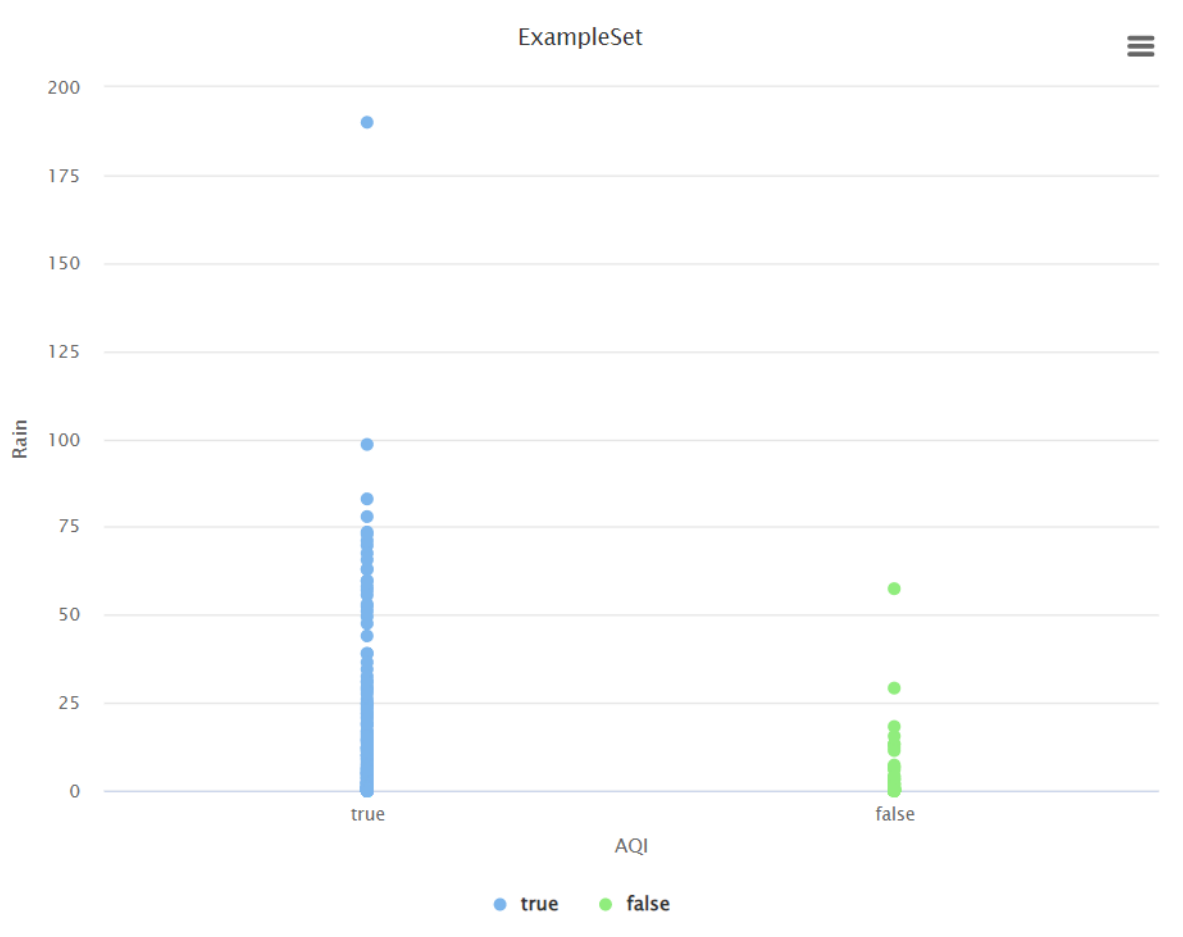


Figure 30: the imbalanced dataset, plots AQI on x-axis and precipitation on y-axis

8.1.1 F-score

There are some indicators we need to define first:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

The indicator precision means the proportion of correctly predicted positive samples among all samples predicted as positive. While the indicators recall presents the proportion of that were correctly predicted as positive among all actual positive samples. With these two indicators, we can define F1-Score F_β or use F_β to adjust the ratio of precision and recall. Note that β is the ratio of precision and recall.

$$F1 - Score = \frac{2 * P * R}{P + R}$$

$$F_{\beta} = \frac{(1 + \beta^2) * PR}{P * \beta^2 + R}$$

8.1.2 Precision-Recall Curves

Precision-Recall Curve, we often call it PR curve, plots recall on the x-axis and precision on the y-axis. Each point are plot with different value of recall and precision. In general, the higher the precision and recall, the better the model performance. Thus, we define AUPRC (Area Under PR curve) to evaluate the model's performance by calculating the area under the PR curve.

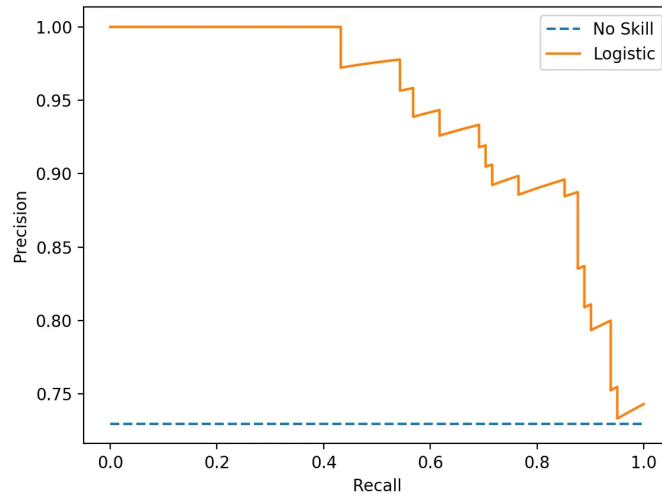


Figure 31: PR curve and AUPRC

8.1.3 ROC Curves

The full name of ROC is Receiver Operator Characteristic Curve. To compute the graph of ROC curve, we first define false positive rate (FPR) and true positive rate (TPR):

$$FPR = \frac{FP}{TN + FP}$$

$$TPR = \frac{FN}{TP + FN}$$

Also, we define the threshold for the model. If the model predicts a data with a probability more than the threshold, we consider it as a positive sample. Otherwise, it is treated as a negative sample. With these indicators, we can plot the ROC curve, which uses FPR on the x-axis and TPR on the y-axis. As in the section of PR curve, we define AUC (Area Under ROC curve) to evaluate the model's performance by calculating the area under the ROC curve. Thus, the higher the AUC, the better the model performance. In general, AUC values are interpreted as follows: 0.5-0.6 (failed), 0.6-0.7 (worthless), 0.7-0.8 (poor), 0.8-0.9 (good), > 0.9 (excellent).

These indicators can help us to have a comprehensive understanding of the quality of our model, and to determine how to modify the model.

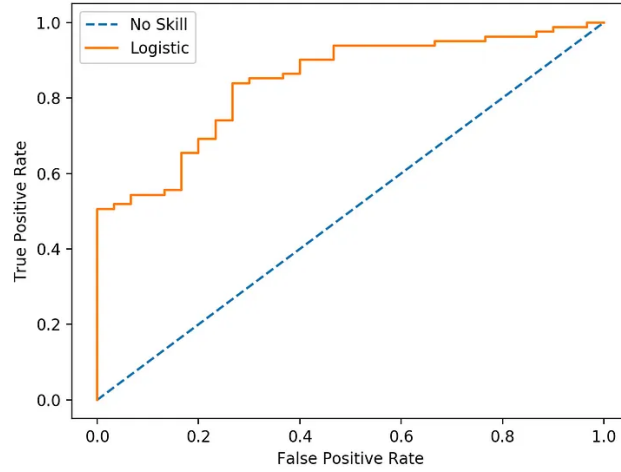


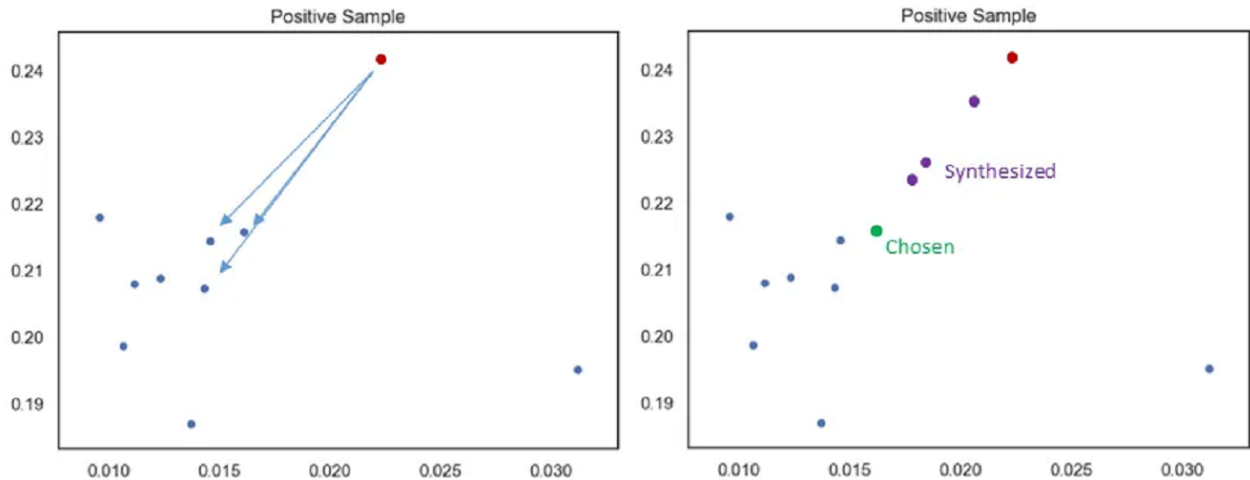
Figure 32: ROC curve and AUC

8.2 Sampling

To solve the problems caused by imbalanced dataset, we need to perform oversampling or undersampling on it. Oversampling is to generate synthetic data points for the minority class, while undersampling is to reduce the number of instances in the majority class. In reality, we can combine two types of method to sample data. Also, there are another thing need to be notice: We need to first split the dataset into a training set and a testing set, and then sample the training set to prevent the model from learning too much noises. Otherwise, the reliability of the test set will decrease. Next, we will introduce two methods.

8.2.1 SMOTE

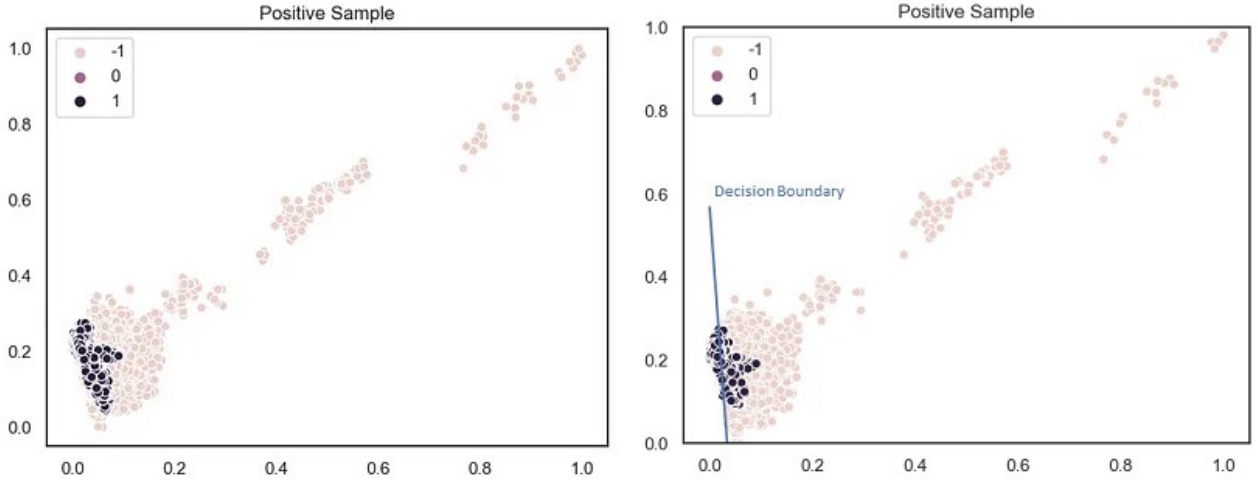
The full name of SMOTE is Synthetic Minority Oversampling Technique. SMOTE is used to conduct oversampling, generating synthetic data points for the minority class. The basic algorithm follows the steps below:



1. Pick up one point of the minority class, call $x_{nearest}$.
2. Define the number of neighbors, k . This means that we need to find the k nearest points to $x_{nearest}$.
3. Randomly choose one point of the k nearest points, called x_{chosen} .
4. Generate new sample point x_{new} by

$$x_{new} = x_{chosen} + (x_{nearest} - x_{chosen}) \cdot \delta; \quad \delta \in [0, 1]$$

5. Apply to all points of the minority class. After applying SMOTE to imbalanced dataset, it would have a better decision boundary.



8.2.2 Tomek Link

Tomek link is a method that used to conduct undersampling, reducing the number of instances in the majority class. We need to be careful not to delete useful data points. Tomek link is a method to find and delete the point that might be considered as noise.

Definition 8.2.3 For an instance pair (x_i, x_j) where $x_i \in S_{max}$ and $x_j \in S_{min}$, if there is no x_k such that $d(x_i, x_k) < d(x_i, x_j)$ or $d(x_j, x_k) < d(x_i, x_j)$, then (x_i, x_j) is a totem link.

After identifying all totem links, the corresponding x_i in each link is then removed.

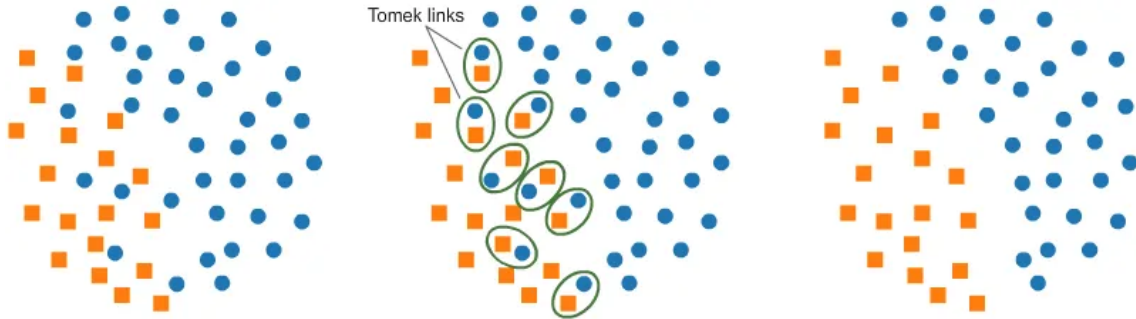


Figure 33: how to find totem link

8.3 Application

We use the extension of RapidMiner Studio, which includes SMOTE method and performance of AUC and AUPRC, to predict AQI again. There are the results:

accuracy: 49.81%

	true false	true true	class precision
pred. false	89	125	41.59%
pred. true	8	43	84.31%
class recall	91.75%	25.60%	

AUC: 0.614、AUPRC: 0.767

Figure 34: 經過 SMOTE 處理、使用雨量預測 AQI

accuracy: 59.62%

	true false	true true	class precision
pred. false	77	87	46.95%
pred. true	20	81	80.20%
class recall	79.38%	48.21%	

AUC: 0.724、AUPRC: 0.800

Figure 35: 經過 SMOTE 處理、使用風速預測 AQI

accuracy: 50.94%

	true false	true true	class precision
pred. false	21	54	28.00%
pred. true	76	114	60.00%
class recall	21.65%	67.86%	

AUC: 0.422、AUPRC: 0.567

Figure 36: 經過 SMOTE 處理、使用風向預測 AQI

accuracy: 65.66%

	true false	true true	class precision
pred. false	36	30	54.55%
pred. true	61	138	69.35%
class recall	37.11%	82.14%	

AUC: 0.738、AUPRC: 0.825

Figure 37: 經過 SMOTE 處理、使用氣溫預測 AQI

accuracy: 43.77%

	true false	true true	class precision
pred. false	55	107	33.95%
pred. true	42	61	59.22%
class recall	56.70%	36.31%	

AUC: 0.500、AUPRC: 0.624

Figure 38: 經過 SMOTE 處理、使用濕度預測 AQI

accuracy: 52.08%

	true false	true true	class precision
pred. false	33	63	34.38%
pred. true	64	105	62.13%
class recall	34.02%	62.50%	

AUC: 0.580、AUPRC: 0.723

Figure 39: 經過 SMOTE 處理、使用紫外線預測 AQI

accuracy: 76.23%

	true false	true true	class precision
pred. false	68	34	66.67%
pred. true	29	134	82.21%
class recall	70.10%	79.76%	

AUC: 0.810、AUPRC: 0.843

Figure 40: 經過 SMOTE 處理、使用前一日 AQI 預測 AQI

8.3.1 Conclusion

We found that the predictions are still not accurate even after processing the dataset with SMOTE. It may be caused by some reasons: Because the original data was not accurate enough, or using only SMOTE could not completely deal with the imbalance in our data.

9 結論

By researching the issue, we first gained a deeper understanding of air pollution and learned many machine learning methods and algorithms. We used one of the methods - SVM to use relevant data to predict AQI values. However, from the operation results it can be seen from the picture that when SVM uses a single factor to predict AQI, the accuracy is only about seventy percent. After discussion, we came to a possible reason: SVM cannot distinguish between right and wrong due to the data, so it guesses all right.

10 附錄

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 41: 1-p

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 42: 2-p-la

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 43: 2-p-m

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 44: 2-p-r

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 45: 2-p-t

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 46: 2-p-uv

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 47: 2-p-wd

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 48: 2-p-ws

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 49: 3-p-m-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 50: 3-p-m-uv

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 51: 3-p-r-la

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 52: 3-p-r-m

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 53: 3-p-r-t

accuracy: 95.09%

	true false	true true	class precision
pred. false	69	3	95.83%
pred. true	10	183	94.82%
class recall	87.34%	98.39%	

Figure 54: 3-p-r-uv

accuracy: 95.09%

	true false	true true	class precision
pred. false	69	3	95.83%
pred. true	10	183	94.82%
class recall	87.34%	98.39%	

Figure 55: 3-p-r-wd

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 56: 3-p-r-ws

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 57: 3-p-t-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 58: 3-p-t-m

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 59: 3-p-t-uv

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 60: 3-p-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 61: 3-p-wd-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 62: 3-p-wd-m

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 63: 3-p-wd-t

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 64: 3-p-wd-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 65: 3-p-ws-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 66: 3-p-ws-m

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 67: 3-p-ws-t

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 68: 3-p-ws-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 69: 3-p-ws-wd

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 70: 4-p-m-uv-la

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 71: 4-p-r-m-la

accuracy: 95.09%

	true false	true true	class precision
pred. false	69	3	95.83%
pred. true	10	183	94.82%
class recall	87.34%	98.39%	

Figure 72: 4-p-r-m-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 73: 4-p-r-t-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 74: 4-p-r-t-m

accuracy: 95.85%

	true false	true true	class precision
pred. false	70	2	97.22%
pred. true	9	184	95.34%
class recall	88.61%	98.92%	

Figure 75: 4-p-r-t-uv

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 76: 4-p-r-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 77: 4-p-r-wd-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 78: 4-p-m-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 79: 4-p-r-wd-t

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 80: 4-p-r-wd-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 81: 4-p-r-ws-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 82: 4-p-r-ws-m

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 83: 4-p-r-ws-t

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 84: 4-p-r-ws-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 85: 4-p-r-ws-wd

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 86: 4-p-t-m-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	70	5	93.33%
pred. true	9	181	95.26%
class recall	88.61%	97.31%	

Figure 87: 4-p-t-m-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 92: 4-p-wd-t-m

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 88: 4-p-t-uv-la

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 89: 4-p-wd-m-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 90: 4-p-wd-m-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 91: 4-p-wd-t-la

accuracy: 95.85%

	true false	true true	class precision
pred. false	70	2	97.22%
pred. true	9	184	95.34%
class recall	88.61%	98.92%	

Figure 93: 4-p-wd-t-uv

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 94: 4-p-wd-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 95: 4-p-ws-m-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	67	3	95.71%
pred. true	12	183	93.85%
class recall	84.81%	98.39%	

Figure 96: 4-p-ws-m-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 97: 4-p-ws-t-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 98: 4-p-ws-t-m

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 99: 4-p-ws-t-uv

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 100: 4-p-ws-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 101: 4-p-ws-wd-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 102: 4-p-ws-wd-m

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 103: 4-p-ws-wd-t

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 104: 4-p-ws-wd-uv

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 105: 5-p-r-m-uv-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 106: 5-p-r-t-m-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	69	5	93.24%
pred. true	10	181	94.76%
class recall	87.34%	97.31%	

Figure 107: 5-p-r-t-m-uv

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 108: 5-p-r-t-uv-la

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 109: 5-p-r-wd-m-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 110: 5-p-r-wd-m-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 111: 5-p-r-wd-t-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 112: 5-p-r-wd-t-m

accuracy: 95.47%

	true false	true true	class precision
pred. false	70	3	95.89%
pred. true	9	183	95.31%
class recall	88.61%	98.39%	

Figure 113: 5-p-r-wd-t-uv

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 114: 5-p-r-wd-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	69	4	94.52%
pred. true	10	182	94.79%
class recall	87.34%	97.85%	

Figure 115: 5-p-r-ws-m-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 116: 5-p-r-ws-m-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 117: 5-p-r-ws-t-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 118: 5-p-r-ws-t-m

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 119: 5-p-r-ws-t-uv

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 120: 5-p-r-ws-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 121: 5-p-r-ws-wd-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 122: 5-p-r-ws-wd-m

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 123: 5-p-r-ws-wd-t

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 124: 5-p-r-ws-wd-uv

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 125: 5-p-t-m-uv-la

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 126: 5-p-wd-m-uv-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 127: 5-p-wd-t-m-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	69	5	93.24%
pred. true	10	181	94.76%
class recall	87.34%	97.31%	

Figure 128: 5-p-wd-t-m-uv

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 129: 5-p-wd-t-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 130: 5-p-ws-m-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 131: 5-p-ws-t-m-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 132: 5-p-ws-t-m-uv

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 133: 5-p-ws-t-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 134: 5-p-ws-wd-m-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	67	3	95.71%
pred. true	12	183	93.85%
class recall	84.81%	98.39%	

Figure 135: 5-p-ws-wd-m-uv

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 136: 5-p-ws-wd-t-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 137: 5-p-ws-wd-t-m

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 138: 5-p-ws-wd-t-uv

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 139: 5-p-ws-wd-uv-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	69	5	93.24%
pred. true	10	181	94.76%
class recall	87.34%	97.31%	

Figure 140: 6-p-r-t-m-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 141: 6-p-r-wd-m-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 142: 6-p-r-wd-t-m-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	69	5	93.24%
pred. true	10	181	94.76%
class recall	87.34%	97.31%	

Figure 143: 6-p-r-wd-t-m-uv

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 144: 6-p-r-wd-t-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 145: 6-p-r-ws-m-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 146: 6-p-r-ws-t-m-la

accuracy: 93.96%

	true false	true true	class precision
pred. false	68	5	93.15%
pred. true	11	181	94.27%
class recall	86.08%	97.31%	

Figure 147: 6-p-r-ws-t-m-uv

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 148: 6-p-r-ws-t-uv-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 149: 6-p-r-ws-wd-m-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	67	3	95.71%
pred. true	12	183	93.85%
class recall	84.81%	98.39%	

Figure 150: 6-p-r-ws-wd-m-uv

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 151: 6-p-r-ws-wd-t-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 152: 6-p-r-ws-wd-t-m

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 153: 6-p-r-ws-wd-t-uv

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 154: 6-p-r-ws-wd-uv-la

accuracy: 95.85%

	true false	true true	class precision
pred. false	70	2	97.22%
pred. true	9	184	95.34%
class recall	88.61%	98.92%	

Figure 155: 6-p-wd-t-m-uv-la

accuracy: 95.47%

	true false	true true	class precision
pred. false	69	2	97.18%
pred. true	10	184	94.85%
class recall	87.34%	98.92%	

Figure 156: 6-p-ws-t-m-uv-la

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 157: 6-p-ws-wd-m-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 158: 6-p-ws-wd-t-m-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 159: 6-p-ws-wd-t-m-uv

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 160: 6-p-ws-wd-t-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	69	4	94.52%
pred. true	10	182	94.79%
class recall	87.34%	97.85%	

Figure 161: 7-p-r-wd-t-m-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 162: 7-p-r-ws-t-m-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 163: 7-p-r-ws-wd-m-uv-la

accuracy: 94.34%

	true false	true true	class precision
pred. false	68	4	94.44%
pred. true	11	182	94.30%
class recall	86.08%	97.85%	

Figure 164: 7-p-r-ws-wd-t-m-la

accuracy: 93.96%

	true false	true true	class precision
pred. false	68	5	93.15%
pred. true	11	181	94.27%
class recall	86.08%	97.31%	

class recall

Figure 165: 7-p-r-ws-wd-t-m-uv

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 166: 7-p-r-ws-wd-t-uv-la

accuracy: 95.09%

	true false	true true	class precision
pred. false	68	2	97.14%
pred. true	11	184	94.36%
class recall	86.08%	98.92%	

Figure 167: 7-p-ws-wd-t-m-uv-la

accuracy: 94.72%

	true false	true true	class precision
pred. false	68	3	95.77%
pred. true	11	183	94.33%
class recall	86.08%	98.39%	

Figure 168: 8-p-r-ws-wd-t-m-uv-la

accuracy: 80.75%

	true false	true true	class precision
pred. false	38	10	79.17%
pred. true	41	176	81.11%
class recall	48.10%	94.62%	

Figure 169: 1-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 170: 1-m

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 171: 1-t

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 172: 1-uv

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 173: 1-r

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 174: 1-ws

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 175: 1-wd

accuracy: 80.75%

	true false	true true	class precision
pred. false	38	10	79.17%
pred. true	41	176	81.11%
class recall	48.10%	94.62%	

Figure 176: 2-m-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 177: 2-m-uv

accuracy: 81.51%

	true false	true true	class precision
pred. false	42	12	77.78%
pred. true	37	174	82.46%
class recall	53.16%	93.55%	

Figure 178: 2-r-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 179: 2-r-m

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 180: 2-r-t

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 181: 2-r-uv

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 182: 2-r-wd

accuracy: 70.19%

	true false	true true	class precision
pred. false	35	35	50.00%
pred. true	44	151	77.44%
class recall	44.30%	81.18%	

Figure 183: 2-r-ws

accuracy: 80.00%

	true false	true true	class precision
pred. false	39	13	75.00%
pred. true	40	173	81.22%
class recall	49.37%	93.01%	

Figure 184: 2-t-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 185: 2-t-m

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 186: 2-t-uv

accuracy: 80.38%

	true false	true true	class precision
pred. false	38	11	77.55%
pred. true	41	175	81.02%
class recall	48.10%	94.09%	

Figure 187: 2-uv-la

accuracy: 80.38%

	true false	true true	class precision
pred. false	39	12	76.47%
pred. true	40	174	81.31%
class recall	49.37%	93.55%	

Figure 188: 2-wd-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 189: 2-wd-m

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 190: 2-wd-t

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 191: 2-wd-uv

accuracy: 81.13%

	true false	true true	class precision
pred. false	41	12	77.36%
pred. true	38	174	82.08%
class recall	51.90%	93.55%	

Figure 192: 2-ws-la

accuracy: 72.83%

	true false	true true	class precision
pred. false	19	12	61.29%
pred. true	60	174	74.36%
class recall	24.05%	93.55%	

Figure 193: 2-ws-m

accuracy: 75.85%

	true false	true true	class precision
pred. false	34	19	64.15%
pred. true	45	19 107	78.77%
class recall	43.04%	89.78%	

Figure 194: 2-ws-t

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 195: 2-ws-uv

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 196: 2-ws-wd

accuracy: 80.38%

	true false	true true	class precision
pred. false	37	10	78.72%
pred. true	42	176	80.73%
class recall	46.84%	94.62%	

Figure 197: 3-m-uv-la

accuracy: 81.51%

	true false	true true	class precision
pred. false	45	15	75.00%
pred. true	34	171	83.41%
class recall	56.96%	91.94%	

Figure 198: 3-r-m-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 199: 3-r-m-uv

accuracy: 81.51%

	true false	true true	class precision
pred. false	45	15	75.00%
pred. true	34	171	83.41%
class recall	56.96%	91.94%	

Figure 200: 3-r-t-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 201: 3-r-t-m

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 202: 3-r-t-uv

accuracy: 81.13%

	true false	true true	class precision
pred. false	42	13	76.36%
pred. true	37	173	82.38%
class recall	53.16%	93.01%	

Figure 203: 3-r-uv-la

accuracy: 80.75%

	true false	true true	class precision
pred. false	41	13	75.93%
pred. true	38	173	81.99%
class recall	51.90%	93.01%	

Figure 204: 3-r-wd-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 205: 3-r-wd-m

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 206: 3-r-wd-t

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 207: 3-r-wd-uv

accuracy: 83.02%

	true false	true true	class precision
pred. false	47	13	78.33%
pred. true	32	173	84.39%
class recall	59.49%	93.01%	

Figure 208: 3-r-ws-la

accuracy: 72.45%

	true false	true true	class precision
pred. false	28	22	56.00%
pred. true	51	164	76.28%
class recall	35.44%	88.17%	

Figure 209: 3-r-ws-m

accuracy: 76.98%

	true false	true true	class precision
pred. false	39	21	65.00%
pred. true	40	165	80.49%
class recall	49.37%	88.71%	

Figure 210: 3-r-ws-t

accuracy: 74.34%

	true false	true true	class precision
pred. false	20	9	68.97%
pred. true	59	177	75.00%
class recall	25.32%	95.16%	

Figure 211: 3-r-ws-uv

accuracy: 70.19%

	true false	true true	class precision
pred. false	24	24	50.00%
pred. true	55	162	74.65%
class recall	30.38%	87.10%	

Figure 212: 3-r-ws-wd

accuracy: 81.13%

	true false	true true	class precision
pred. false	42	13	76.36%
pred. true	37	173	82.38%
class recall	53.16%	93.01%	

Figure 213: 3-t-m-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 214: 3-t-m-uv

accuracy: 81.51%

	true false	true true	class precision
pred. false	44	14	75.86%
pred. true	35	172	83.09%
class recall	55.70%	92.47%	

Figure 215: 3-t-uv-la

accuracy: 80.38%

	true false	true true	class precision
pred. false	39	12	76.47%
pred. true	40	174	81.31%
class recall	49.37%	93.55%	

Figure 216: 3-wd-m-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 217: 3-wd-m-uv

accuracy: 80.75%

	true false	true true	class precision
pred. false	42	14	75.00%
pred. true	37	172	82.30%
class recall	53.16%	92.47%	

Figure 218: 3-wd-t-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 219: 3-wd-t-m

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 220: 3-wd-t-uv

accuracy: 80.00%

	true false	true true	class precision
pred. false	38	12	76.00%
pred. true	41	174	80.93%
class recall	48.10%	93.55%	

Figure 221: 3-wd-uv-la

accuracy: 81.51%

	true false	true true	class precision
pred. false	42	12	77.78%
pred. true	37	174	82.46%
class recall	53.16%	93.55%	

Figure 222: 3-ws-m-la

accuracy: 74.34%

	true false	true true	class precision
pred. false	13	2	86.67%
pred. true	66	184	73.60%
class recall	16.46%	98.92%	

Figure 223: 3-m-uv

accuracy: 81.89%

	true false	true true	class precision
pred. false	48	17	73.85%
pred. true	31	169	84.50%
class recall	60.76%	90.86%	

Figure 224: 3-ws-t-la

accuracy: 78.49%

	true false	true true	class precision
pred. false	33	11	75.00%
pred. true	46	175	79.19%
class recall	41.77%	94.09%	

Figure 225: 3-ws-t-m

accuracy: 77.36%

	true false	true true	class precision
pred. false	36	17	67.92%
pred. true	43	169	79.72%
class recall	45.57%	90.86%	

Figure 226: 3-ws-t-uv

accuracy: 77.36%

	true false	true true	class precision
pred. false	36	17	67.92%
pred. true	43	169	79.72%
class recall	45.57%	90.86%	

Figure 227: 3-ws-t-uv

accuracy: 81.51%

	true false	true true	class precision
pred. false	42	12	77.78%
pred. true	37	174	82.46%
class recall	53.16%	93.55%	

Figure 228: 3-ws-uv-la

accuracy: 81.13%

	true false	true true	class precision
pred. false	41	12	77.36%
pred. true	38	174	82.08%
class recall	51.90%	93.55%	

Figure 229: 3-ws-wd-la

accuracy: 73.58%

	true false	true true	class precision
pred. false	16	7	69.57%
pred. true	63	179	73.97%
class recall	20.25%	96.24%	

Figure 230: 3-ws-wd-m

accuracy: 76.23%

	true false	true true	class precision
pred. false	33	17	66.00%
pred. true	46	169	78.60%
class recall	41.77%	90.86%	

Figure 231: 3-ws-wd-t

accuracy: 81.51%

	true false	true true	class precision
pred. false	41	11	78.85%
pred. true	38	175	82.16%
class recall	51.90%	94.09%	

Figure 232: 4-r-m-uv-la

accuracy: 81.51%

	true false	true true	class precision
pred. false	45	15	75.00%
pred. true	34	171	83.41%
class recall	56.96%	91.94%	

Figure 233: 4-r-t-m-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 234: 4-r-t-m-uv

accuracy: 82.26%

	true false	true true	class precision
pred. false	48	16	75.00%
pred. true	31	170	84.58%
class recall	60.76%	91.40%	

Figure 235: 4-r-t-uv-la

accuracy: 80.38%

	true false	true true	class precision
pred. false	40	13	75.47%
pred. true	39	173	81.60%
class recall	50.63%	93.01%	

Figure 236: 4-r-wd-m-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 237: 4-r-wd-m-uv

accuracy: 81.51%

	true false	true true	class precision
pred. false	47	17	73.44%
pred. true	32	169	84.08%
class recall	59.49%	90.86%	

Figure 238: 4-r-wd-t-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 239: 4-r-wd-t-m

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 240: 4-r-wd-t-uv

accuracy: 80.38%

	true false	true true	class precision
pred. false	42	15	73.68%
pred. true	37	171	82.21%
class recall	53.16%	91.94%	

Figure 241: 4-r-wd-uv-la

accuracy: 81.13%

	true false	true true	class precision
pred. false	43	14	75.44%
pred. true	36	172	82.69%
class recall	54.43%	92.47%	

Figure 242: 4-r-ws-m-la

accuracy: 75.85%

	true false	true true	class precision
pred. false	19	4	82.61%
pred. true	60	182	75.21%
class recall	24.05%	97.85%	

Figure 243: 4-r-ws-m-uv

accuracy: 83.02%

	true false	true true	class precision
pred. false	51	17	75.00%
pred. true	28	169	85.79%
class recall	64.56%	90.86%	

Figure 244: 4-r-ws-t-la

accuracy: 83.02%

	true false	true true	class precision
pred. false	51	17	75.00%
pred. true	28	169	85.79%
class recall	64.56%	90.86%	

Figure 245: 4-r-ws-t-la

accuracy: 78.87%

	true false	true true	class precision
pred. false	34	11	75.56%
pred. true	45	175	79.55%
class recall	43.04%	94.09%	

Figure 246: 4-r-ws-t-m

accuracy: 78.11%

	true false	true true	class precision
pred. false	38	17	69.09%
pred. true	41	169	80.48%
class recall	48.10%	90.86%	

Figure 247: 4-r-ws-t-uv

accuracy: 81.13%

	true false	true true	class precision
pred. false	46	17	73.02%
pred. true	33	169	83.66%
class recall	58.23%	90.86%	

Figure 248: 4-r-ws-uv-la

accuracy: 82.64%

	true false	true true	class precision
pred. false	46	13	77.97%
pred. true	33	173	83.98%
class recall	58.23%	93.01%	

Figure 249: 4-r-ws-wd-la

accuracy: 72.08%

	true false	true true	class precision
pred. false	28	23	54.90%
pred. true	51	163	76.17%
class recall	35.44%	87.63%	

Figure 250: 4-r-ws-wd-m

accuracy: 76.98%

	true false	true true	class precision
pred. false	36	18	66.67%
pred. true	43	168	79.62%
class recall	45.57%	90.32%	

Figure 251: 4-r-ws-wd-t

accuracy: 72.08%

	true false	true true	class precision
pred. false	22	17	56.41%
pred. true	57	169	74.78%
class recall	27.85%	90.86%	

Figure 252: 4-r-ws-wd-uv

accuracy: 81.51%

	true false	true true	class precision
pred. false	44	14	75.86%
pred. true	35	172	83.09%
class recall	55.70%	92.47%	

Figure 253: 4-t-m-uv-la

accuracy: 80.38%

	true false	true true	class precision
pred. false	39	12	76.47%
pred. true	40	174	81.31%
class recall	49.37%	93.55%	

Figure 254: 4-wd-m-uv-la

accuracy: 81.51%

	true false	true true	class precision
pred. false	45	15	75.00%
pred. true	34	171	83.41%
class recall	56.96%	91.94%	

Figure 255: 4-wd-t-m-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 256: 4-wd-t-m-uv

accuracy: 80.75%

	true false	true true	class precision
pred. false	44	16	73.33%
pred. true	35	170	82.93%
class recall	55.70%	91.40%	

Figure 257: 4-wd-t-uv-la

accuracy: 80.38%

	true false	true true	class precision
pred. false	40	13	75.47%
pred. true	39	173	81.60%
class recall	50.63%	93.01%	

Figure 258: 4-ws-m-uv-la

accuracy: 83.02%

	true false	true true	class precision
pred. false	46	12	79.31%
pred. true	33	174	84.06%
class recall	58.23%	93.55%	

Figure 259: 4-ws-t-m-la

accuracy: 79.25%

	true false	true true	class precision
pred. false	38	14	73.08%
pred. true	41	172	80.75%
class recall	48.10%	92.47%	

Figure 260: 4-ws-t-m-uv

accuracy: 82.26%

	true false	true true	class precision
pred. false	48	16	75.00%
pred. true	31	170	84.58%
class recall	60.76%	91.40%	

Figure 261: 4-ws-t-uv-la

accuracy: 81.13%

	true false	true true	class precision
pred. false	42	13	76.36%
pred. true	37	173	82.38%
class recall	53.16%	93.01%	

Figure 262: 4-ws-wd-m-la

accuracy: 73.96%

	true false	true true	class precision
pred. false	21	11	65.62%
pred. true	58	175	75.11%
class recall	26.58%	94.09%	

Figure 263: 4-ws-wd-m-uv

accuracy: 81.13%

	true false	true true	class precision
pred. false	48	19	71.64%
pred. true	31	167	84.34%
class recall	60.76%	89.78%	

Figure 264: 4-ws-wd-m-uv

accuracy: 81.13%

	true false	true true	class precision
pred. false	48	19	71.64%
pred. true	31	167	84.34%
class recall	60.76%	89.78%	

Figure 265: 4-ws-wd-t-la

accuracy: 79.25%

	true false	true true	class precision
pred. false	37	13	74.00%
pred. true	42	173	80.47%
class recall	46.84%	93.01%	

Figure 266: 4-ws-wd-t-m

accuracy: 77.74%

	true false	true true	class precision
pred. false	36	16	69.23%
pred. true	43	170	79.81%
class recall	45.57%	91.40%	

Figure 267: 4-ws-wd-t-uv

accuracy: 80.75%

	true false	true true	class precision
pred. false	40	12	76.92%
pred. true	39	174	81.69%
class recall	50.63%	93.55%	

Figure 268: 4-ws-wd-uv-la

accuracy: 80.75%

	true false	true true	class precision
pred. false	47	19	71.21%
pred. true	32	167	83.92%
class recall	59.49%	89.78%	

Figure 269: 5-r-t-m-uv-la

accuracy: 80.00%

	true false	true true	class precision
pred. false	41	15	73.21%
pred. true	38	171	81.82%
class recall	51.90%	91.94%	

Figure 270: 5-r-wd-m-uv-la

accuracy: 81.13%

	true false	true true	class precision
pred. false	45	16	73.77%
pred. true	34	170	83.33%
class recall	56.96%	91.40%	

Figure 271: 5-r-wd-t-m-la

accuracy: 70.19%

	true false	true true	class precision
pred. false	0	0	0.00%
pred. true	79	186	70.19%
class recall	0.00%	100.00%	

Figure 272: 5-r-wd-t-m-uv

accuracy: 81.51%

	true false	true true	class precision
pred. false	47	17	73.44%
pred. true	32	169	84.08%
class recall	59.49%	90.86%	

Figure 273: 5-r-wd-t-uv-la

accuracy: 82.64%

	true false	true true	class precision
pred. false	46	13	77.97%
pred. true	33	173	83.98%
class recall	58.23%	93.01%	

Figure 274: 5-r-ws-m-uv-la

accuracy: 84.15%

	true false	true true	class precision
pred. false	48	11	81.36%
pred. true	31	175	84.95%
class recall	60.76%	94.09%	

Figure 275: 5-r-ws-t-m-la

accuracy: 79.25%

	true false	true true	class precision
pred. false	39	15	72.22%
pred. true	40	171	81.04%
class recall	49.37%	91.94%	

Figure 276: 5-r-ws-t-m-uv

accuracy: 83.77%

	true false	true true	class precision
pred. false	51	15	77.27%
pred. true	28	171	85.93%
class recall	64.56%	91.94%	

Figure 277: 5-r-ws-t-uv-la

accuracy: 82.64%

	true false	true true	class precision
pred. false	46	13	77.97%
pred. true	33	173	83.98%
class recall	58.23%	93.01%	

Figure 278: 5-r-ws-m-uv-la

accuracy: 84.15%

	true false	true true	class precision
pred. false	48	11	81.36%
pred. true	31	175	84.95%
class recall	60.76%	94.09%	

Figure 279: 5-r-ws-t-m-la

accuracy: 79.25%

	true false	true true	class precision
pred. false	39	15	72.22%
pred. true	40	171	81.04%
class recall	49.37%	91.94%	

Figure 280: 5-r-ws-t-m-uv

accuracy: 83.77%

	true false	true true	class precision
pred. false	51	15	77.27%
pred. true	28	171	85.93%
class recall	64.56%	91.94%	

Figure 281: 5-r-ws-t-uv-la

accuracy: 81.13%

	true false	true true	class precision
pred. false	43	14	75.44%
pred. true	36	172	82.69%
class recall	54.43%	92.47%	

Figure 282: 5-r-ws-wd-m-la

accuracy: 73.96%

	true false	true true	class precision
pred. false	22	12	64.71%
pred. true	57	174	75.32%
class recall	27.85%	93.55%	

Figure 283: 5-r-ws-wd-m-uv

accuracy: 83.02%

	true false	true true	class precision
pred. false	51	17	75.00%
pred. true	28	169	85.79%
class recall	64.56%	90.86%	

Figure 284: 5-r-ws-wd-t-la

accuracy: 79.25%

	true false	true true	class precision
pred. false	37	13	74.00%
pred. true	42	173	80.47%
class recall	46.84%	93.01%	

Figure 285: 5-r-ws-wd-t-m

accuracy: 78.49%

	true false	true true	class precision
pred. false	37	15	71.15%
pred. true	42	171	80.28%
class recall	46.84%	91.94%	

Figure 286: 5-r-ws-wd-t-uv

accuracy: 81.13%

	true false	true true	class precision
pred. false	46	17	73.02%
pred. true	33	169	83.66%
class recall	58.23%	90.86%	

Figure 287: 5-r-ws-wd-uv-la

accuracy: 81.13%

	true false	true true	class precision
pred. false	44	15	74.58%
pred. true	35	171	83.01%
class recall	55.70%	91.94%	

Figure 288: 5-wd-t-m-uv-la

accuracy: 82.64%

	true false	true true	class precision
pred. false	46	13	77.97%
pred. true	33	173	83.98%
class recall	58.23%	93.01%	

Figure 289: 5-ws-t-m-uv-la

accuracy: 80.38%

	true false	true true	class precision
pred. false	39	12	76.47%
pred. true	40	174	81.31%
class recall	49.37%	93.55%	

Figure 290: 5-ws-wd-m-uv-la

accuracy: 82.26%

	true false	true true	class precision
pred. false	45	13	77.59%
pred. true	34	173	83.57%
class recall	56.96%	93.01%	

Figure 291: 5-ws-wd-t-m-la

accuracy: 78.87%

	true false	true true	class precision
pred. false	38	15	71.70%
pred. true	41	171	80.66%
class recall	48.10%	91.94%	

Figure 292: 5-ws-wd-t-m-uv

accuracy: 83.02%

	true false	true true	class precision
pred. false	48	14	77.42%
pred. true	31	172	84.73%
class recall	60.76%	92.47%	

Figure 293: 5-ws-wd-t-uv-la

accuracy: 81.51%

	true false	true true	class precision
pred. false	46	16	74.19%
pred. true	33	170	83.74%
class recall	58.23%	91.40%	

Figure 294: 6-r-wd-t-m-uv-la

accuracy: 83.40%

	true false	true true	class precision
pred. false	47	12	79.66%
pred. true	32	174	84.47%
class recall	59.49%	93.55%	

Figure 295: 6-r-ws-t-m-uv-la

accuracy: 81.51%

	true false	true true	class precision
pred. false	43	13	76.79%
pred. true	36	173	82.78%
class recall	54.43%	93.01%	

Figure 296: 6-r-ws-wd-m-uv-la

accuracy: 83.02%

	true false	true true	class precision
pred. false	47	13	78.33%
pred. true	32	173	84.39%
class recall	59.49%	93.01%	

Figure 297: 6-r-ws-wd-t-m-la

accuracy: 78.87%

	true false	true true	class precision
pred. false	37	14	72.55%
pred. true	42	172	80.37%
class recall	46.84%	92.47%	

Figure 298: 6-r-ws-wd-t-m-uv

accuracy: 82.64%

	true false	true true	class precision
pred. false	48	15	76.19%
pred. true	31	171	84.65%
class recall	60.76%	91.94%	

Figure 299: 6-r-ws-wd-t-uv-la

accuracy: 81.89%

	true false	true true	class precision
pred. false	44	13	77.19%
pred. true	35	173	83.17%
class recall	55.70%	93.01%	

Figure 300: 6-ws-wd-t-m-uv-la

accuracy: 83.02%

	true false	true true	class precision
pred. false	48	14	77.42%
pred. true	31	172	84.73%
class recall	60.76%	92.47%	

Figure 301: 7-r-ws-wd-t-m-uv-la

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